

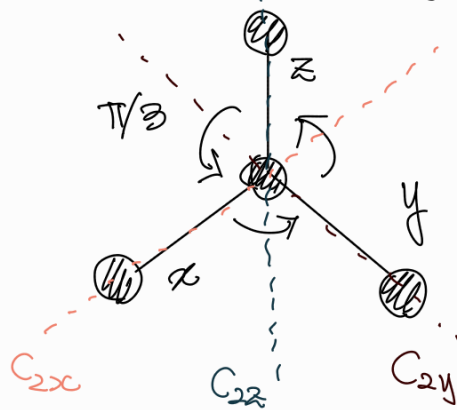
Solving an exercise with the paper: "Fully Generalized Spin Models with strain Effects of Kitaev Spin Liquid Candidate Materials"

In the k -sector, there are only three terms remained:

$$K = \{ J_x^{xx}, J_y^{yy}, J_z^{zz} \}$$

There six elements in D_3 group = $\{ e, C_3, C_3^2, C_{2x}, C_{2y}, C_{2z} \}$.

C_3 rotates $x \rightarrow y, y \rightarrow z, z \rightarrow x$ "bond" in Honeycomb Kitaev model.



There is one permutation structure for C_3 :

$$J_x^{xx} \rightarrow J_y^{yy} \rightarrow J_z^{zz}$$

However, C_2 leaves the bond direction invariant: $J_z^{zz} \leftrightarrow J_z^{zz}$, etc.

Therefore, the character for each operator is given by: $\chi_K(I) = 3, \chi_K(C_3) = 0, \chi_K(C_2) = 1$.

One can decompose the reducible representation using character table:

D_3	I	C_3	C_2
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0

From the great orthogonality theorem,

$$n_\mu = \frac{1}{|G|} \sum_g N_g \chi_\Gamma(g) \chi_\mu(g)$$

where μ denotes irrep., $|G|$ is the order of group (e.g. D_3), g is the class, χ_Γ is the character of reducible repr., and $\chi_\mu(g)$ is the character of irrep, each number of irrep reads:

① $\mu = A_1$.

$$n_{A_1} = \frac{1}{6} \left[\overbrace{1 \times 3 \times 1}^I + \overbrace{2 \times 0 \times 1}^{C_3} + \overbrace{3 \times 1 \times 1}^{C_2} \right] = 1.$$

② $\mu = A_2$.

$$n_{A_2} = \frac{1}{6} \left[1 \times 3 \times 1 + 2 \times 0 \times 1 + 3 \times 1 \times (-1) \right] = 0.$$

② $\mu = E$.

$$n_E = \frac{1}{6} [1 \times 3 \times 2 + 2 \times 0 \times (-1) + 3 \times 1 \times 0] = 1.$$

Therefore,

$$\Gamma = A_1 \oplus E$$

A_1 gives the most symmetric linear combinations, because it should be even for C_3 and C_2 . Its form is determined by:

$$S_k(A_1) = J_x^{xx} + J_y^{yy} + J_z^{zz}$$

This is isotropic Kitaev operator.

Q. How did one get this result?

The projection operator of μ irrep is written as:

$$P_\mu = \frac{d_\mu}{|G|} \sum_g \chi_\mu^*(g) R(g)$$

where d_μ is the dimension of irrep. and $R(g)$ is the representation matrix of the class g . For D_3 , some $R(g)$ can be written as:

$$R(C_3) = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, R(C_{2z}) = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

For A_1 irrep., the projection operator is written as:

$$\begin{aligned} P_{A_1} &= \frac{1}{6} [R(E) + R(C_3) + R(C_3^2) + R(C_{2z}) + R(C_{2x}) + R(C_{2y})] \\ &= \frac{1}{6} \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \end{aligned}$$

The operation on an arbitrary vector $\vec{r} = (x, y, z)^T$ gives

$$P_{A_1} \vec{r} = \frac{x+y+z}{3} (1 \ 1 \ 1)^T$$

Note that $P_{A_1} \vec{r}$ is proportional to $(1 \ 1 \ 1)^T$. This indicates A_1 is isotropic and P_{A_1} only gives isotropic components $J_x^{xx} + J_y^{yy} + J_z^{zz}$.

For E irrep., the projection operator is given by:

$$\begin{aligned}
 P_E &= \frac{2}{6} [2R(I) - R(C_3) - R(C_3^2)] \\
 &= \frac{1}{3} \left[\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \right] \\
 &= I - P_{A_1}
 \end{aligned}$$

The vector multiplied by P_E : $P_E \vec{r}$ must be orthogonal to the A_1 component $(1\ 1\ 1)^T$. It leads to the following relation:

$$\begin{aligned}
 (1\ 1\ 1) P_E \begin{pmatrix} x \\ y \\ z \end{pmatrix} &= (1\ 1\ 1) \begin{bmatrix} 2/3 & -1/3 & -1/3 \\ -1/3 & 2/3 & -1/3 \\ -1/3 & -1/3 & 2/3 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} \\
 &= \frac{1}{3}(x+y+z) = 0, \quad \boxed{\therefore x+y+z=0}
 \end{aligned}$$

When $G_0 \rightarrow G_1$, C_3 is broken and C_{2z} remains. The paper discusses $(\vec{S}^{K(E)})_a$ and $(\vec{S}^{K(E)})_b$ which span the E-representation. This implies that the paper sets C_{2z} as the action in E-subspace and one only needs to choose x, y , and z that makes C_{2z} -even or odd. (to see selection rule for C_{2z})
Such combination is given by:

$$e_a = \frac{1}{\sqrt{2}}(1, -1, 0), \quad e_b = \frac{1}{\sqrt{6}}(1, 1, -2)$$

These vectors are eigenvectors that give ± 1 eigenvalues from $R(C_{2z})$. This allows only the following combinations:

$$S^{K(E)} = \left[\frac{1}{\sqrt{2}}(J_x^{xx} - J_y^{yy}), \frac{1}{\sqrt{6}}(J_x^{xx} + J_y^{yy} - 2J_z^{zz}) \right]$$

* Since I set $J_\beta^{\alpha\alpha} = \sum_{\langle j,k \rangle_\gamma} S_j^\alpha S_k^\alpha$ for simplicity, there is no factor $\frac{1}{2}$. If one writes original one as $\tilde{J}_\beta^{\alpha\alpha}$, it satisfies $\tilde{J}_\beta^{\alpha\alpha} = 2J_\beta^{\alpha\alpha}$ from my notation.